Gemini Notebook

The API (Stateless Brain)

Stateless, cannot act,
no control flow, no governance.

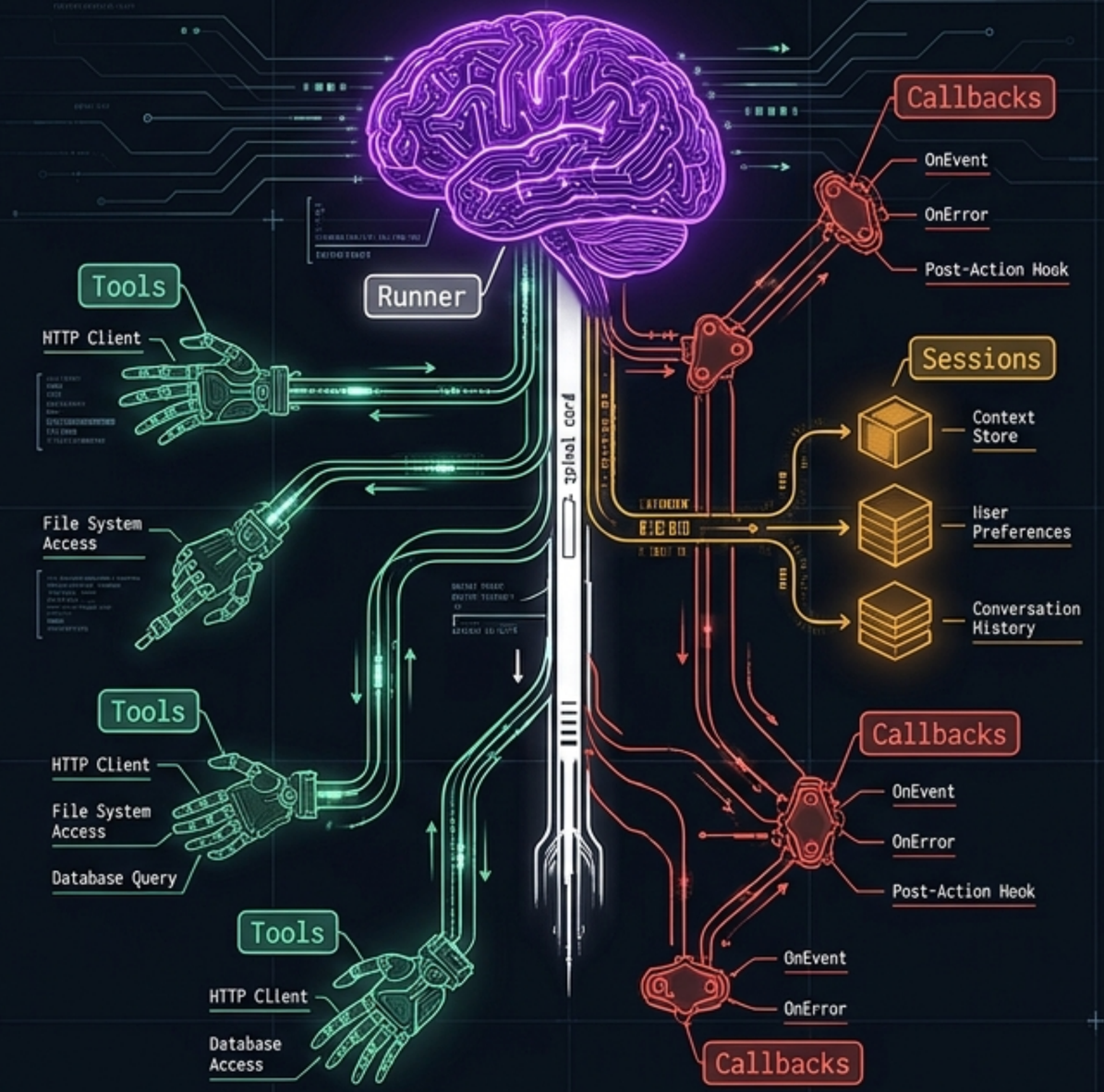
The Agent Theater Litmus Test

Condition 1: Sub-agents have independent tools.	Condition 2: Sub-agents have private data.	Condition 3: Requires an independent verifier role (include_contents='none').	Condition 4: Requires human-gated steps.
Muultiple sub-agents and agents have independent toolsets.	Secure, isolated data vaults, neceessarly data vaults can be coete.	Requires an independent verifier role: (include_contents='none').	Husan-gated steps, since soem that gate has or authorizated steps.

Splitting one prompt into three persona agents + synthesizer costs ~4x tokens and 2x latency for marginal gains. Prove the single call fails first.

Token Usage
Latency ms
Latency ms

The ADK (Full Nervous System)



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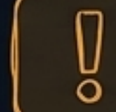
Framework	Mental Model	Best For
ADK	Agent + Runner + SessionService, workflow agents, graph Workflow	GCP/Gemini native, explicit Python control flow, managed sessions & Memory Bank.
LangGraph	StateGraph with nodes, conditional edges, checkpointer	Complex cyclic graphs, time-travel debugging, interrupt() pauses.
CrewAI	Role/goal/backstory teams	Fast prototypes of conversational teams.
PydanticAI	Type-safe agent functions	Schema-first API backends.



SDK ≠ ADK. The Gen AI SDK (google-genai) is a model API client with no persistent sessions, event ledger, or deployment. ADK is the production infrastructure built on top of it.

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Agent
(Stateless)

Holds NO conversation state. One instance serves all users safely.

Contains Model adapter (BaseLlm), instruction, tools.

SessionService
(Stateful)

app_name

user_id

session_id

2. Call model (history + schemas).

1. Load session history.

3. Execute requested tools.

Runner
(Event Loop Engine)

Drives turns via run_async(...)

4. Commit Event (+ state/artifact delta). Must persist before agent continues.

RUNTIME CORE ARCHITECTURE

CYCLICAL EVENT LOOP

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AGENT ROUTING DECISIONS

LlmAgent

Model-driven (ReAct).
Dynamic routing.
Key params:
output_key, planner.

SequentialAgent

Fixed pipeline.
Runs all sub-agents in order.
Does NOT route.

ParallelAgent

Concurrent fan-out.
Independent retrievals.
Must use distinct output_key per branch.

LoopAgent

Repeat until exit.
Key params:
max_iterations,
escalate.

WARNING: Exits SILENTLY when max_Iterations trips. The next step in an outer SequentialAgent will run!

Fix: Record temp:approved = True in the exit tool and gate downstream steps via callbacks.

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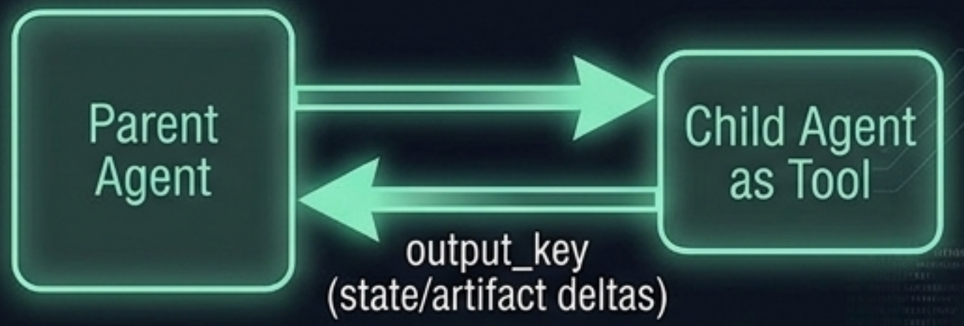
sub_agents (LLM Transfer)

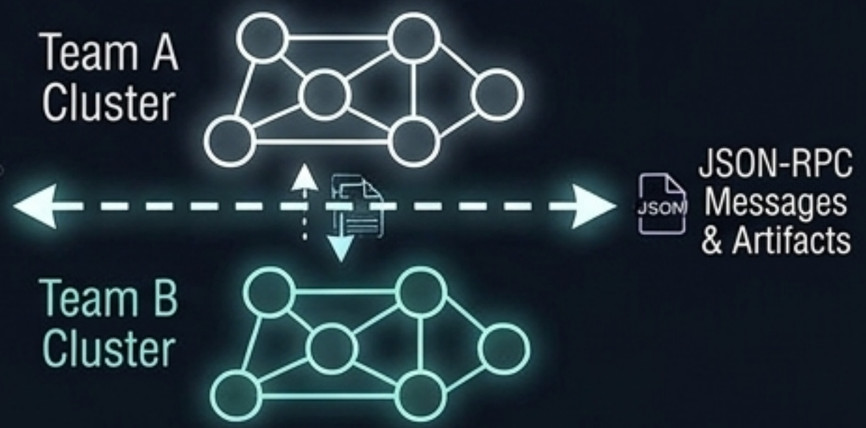
Child takes over chat completely. Model picks target by description. Control: disallow_transfer_to_parent=True forces one-shot utility usage.

AgentTool(agent=...)



Parent calls child like a function. Child forwards state/artifact deltas (output_key) back to parent.

RemoteA2aAgent



Cross-team opaque delegation. Only messages and artifacts cross via JSON-RPC.

Generator-Auditor (Swarm Pattern)

Generator drafts. Auditor verifies. Key detail: Auditor must have temperature=0, include_contents='none', and distinct verification tools to remove self-grading bias.

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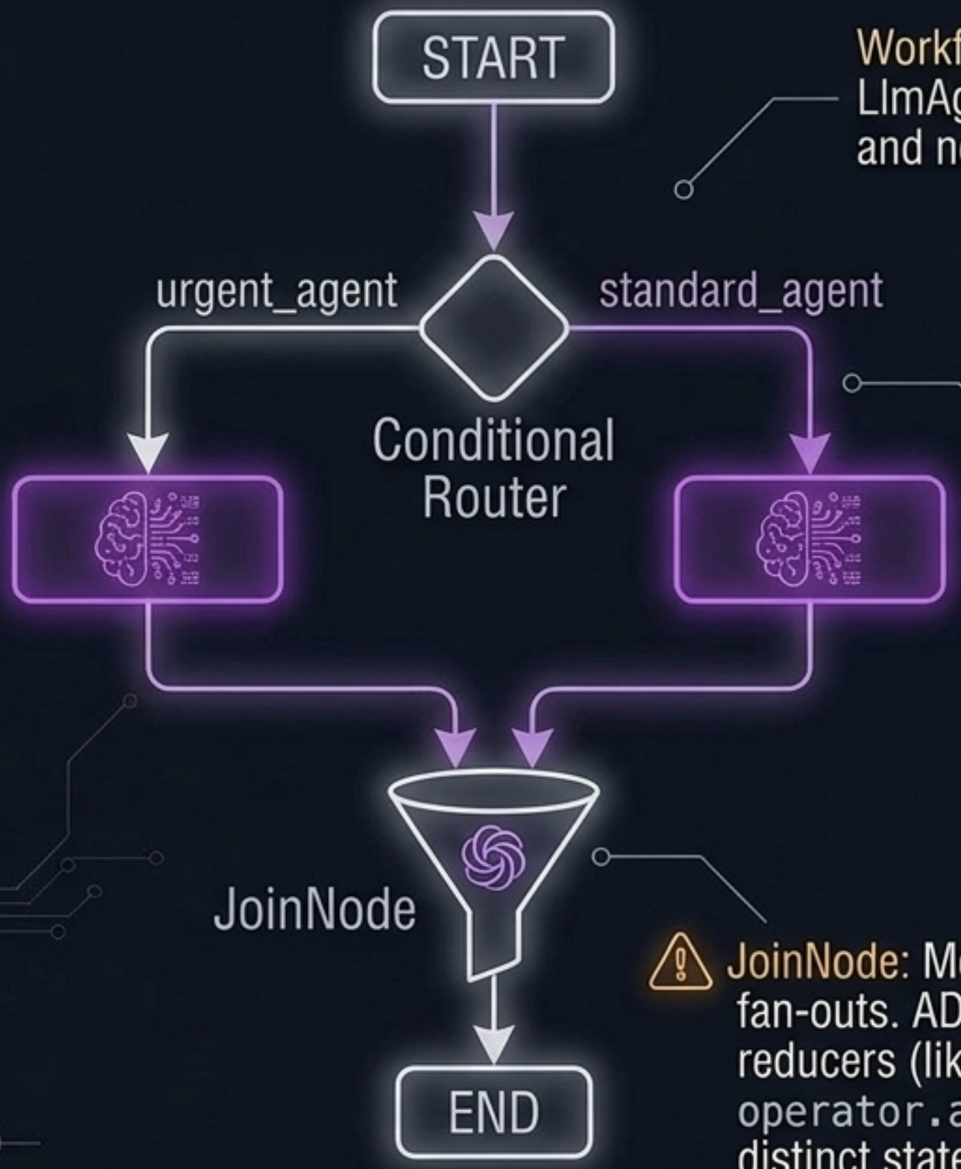

Workflow(edges=[...]): Wires LlmAgents, FunctionNodes, and nested Workflows.

Conditional Edges: Handled by returning Event(route='target') from a node function.

⚠ JoinNode: Merges parallel tuple fan-outs. ADK has NO state reducers (like LangGraph's `operator.add`); you must use distinct state keys.

Planners

- BuiltInPlanner: Uses native thinking tokens (part.thought == True).
- PlanReActPlanner: Forces non-thinking models to emit /*PLANNING*/ → /*ACTION*/ tags.

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2. Function Name: Becomes the tool name. Use domain prefixes to avoid LLM selection collisions.

3. Docstring: This is the UI for the model. Include state limits ('Max 1MB, read-only') so the model plans around constraints.

1. Function Name: Becomes the tool name. Use domain prefixes to avoid LLM selection collisions.

```python
def file_read(path: str) -> dict:
    """Read a UTF-8 text file inside the project
    sandbox (read-only, max 1 MB)."""
    return {"status": "error", "error_message":
    "Not found"}
```

4. Return dict: NEVER raise unhandled exceptions. Return structured errors so the model can retry on the model can retry or escalate. An uncaught exception kills the turn.

Built-in Constraints: Gemini built-ins (`google_search`) run server-side. To mix with custom tools, set `bypass_multi_tools_limit=True`.

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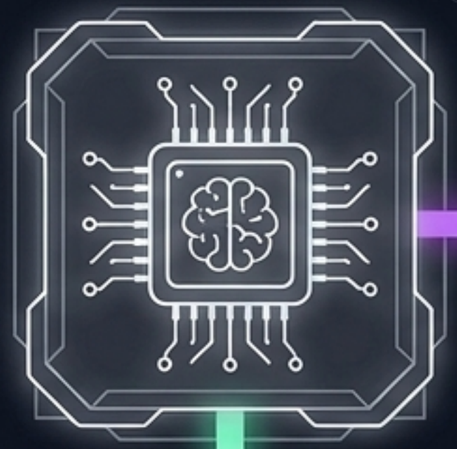

Agent → Tools/Data.

Treat like public APIs. Read-only by default.

stdio

HTTP

Transports: stdio (local subprocess) vs Streamable HTTP (production remote).

min

Security: Use tool_filter (explicit list/predicate) to enforce least privilege.

Vertical Integration: MCP (Model Context Protocol)

Horizontal Integration: A2A (Agent-to-Agent)

Agent → Agent.

JSON-RPC

Opaque peers. Only messages/artifacts artifacts cross via JSON-RPC.

TRUST

Trust: Remote agents run OUTSIDE your callbacks. Validate every returned artifact. Consumed via Agent Cards (/.well-known/agent-card.json).

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L1: Metadata (SkillToolset)

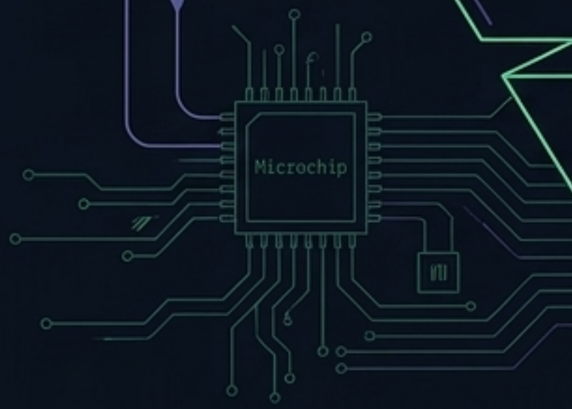
YAML frontmatter name & description (~60 tokens).
YAML frontmatter name & description (~60 tokens).
Loaded eagerly into system prompt or via list_skills.

L2: Instructions

Full SKILL.md body (< 500 lines).
Loaded via load_skill when the
LLM decides it needs it.

L3: Resources

references/, assets/,
scripts/.
Loaded via
run_skill_script.



A malicious SKILL.md is filesystem prompt injection. Provide a sandboxed code_executor when enabling script execution.

Script Execution Rules:

1. Non-interactive (CLI flags only).
2. Self-documenting (--help).
3. Emit structured JSON on stdout.

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`before_agent`: Pre-LLM reality check (Wrong file type? Escalate without spending tokens).

`before_model`: Injection patterns (Regex → small classifier).

`after_model`: Output backstop (Groundedness/citation checks).

`before_tool`: Argument limits (Path traversal checks, SQL injection blocks).

`after_tool`: Indirect injection stripping.

TRAP: Redact PII in `after_tool` or `before_tool`. Doing it in `after_model` is TOO LATE—the raw PII has already entered the model context and the session ledger!

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Level 3: Human Escalation

`require_confirmation=True` on `FunctionTool`
Pauses the graph for human yes/no/payload approval before irreversible actions.
Use `LongRunningFunctionTool` for jobs finishing hours later.

Level 2: Tool Self-Correction

`ReflectAndRetryToolPlugin`
Feeds tool errors back to the LLM so it fixes its own malformed arguments.
Override `extract_error_from_result` if your tools return dicts.

Level 1: Transport

`HttpRetryOptions`
Handles 429/503 network errors automatically.

Resumability

Set `is_resumable=True` in App
Warning: Resumption is at-least-once.
Side-effecting tools require idempotency keys.

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Threat Mitigations

Indirect Injection →	Isolate readers from action tools. Scan in `after_tool_callback`.
Excessive Agency →	tool_filter, read-only credentials, Agent Identity (principal://).
Runaway Cost →	max_llm_calls (default 500), max_iterations.

Code Executors (LlmAgent(code_executor=...))

UnsafeLocalCodeExecutor:
Local host access. NEVER production.

BuiltInCodeExecutor:
Gemini server-side.
Fast, stateless, 30s limit, no internet.

AgentEngineSandboxCodeExecutor:
Managed Cloud runtime.
Stateful across turns (14-day TTL).

GkeCodeExecutor:
GKE Agent Sandbox on gVisor.
Kernel-level isolation, pre-warmed.

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Metrics: Deterministic vs LLM-as-Judge

Deterministic Metrics (Rigid)

Deterministic:
tool_trajectory_avg_score
(EXACT vs IN_ORDER)

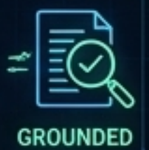
Trap: EXACT fails agents that make harmless extra calls.

Deterministic:
response_match_score
(Lexical/ROUGE-1)

LLM-as-Judge Metrics (Semantic)

LLM:
final_response_match_v2
(Semantic equivalence)

LLM: hallucinations_v1
(Checks groundedness even if the right tool ran)



Observability & Testing:

Observability: Native OpenTelemetry spans (invoke_agent, call_llm). Set *ADK_CAPTURE_MESSAGE_CONTENT_IN_SPANS=false* in prod to keep raw prompts/PII out of Cloud Trace.

Testing: `adk conformance test --mode=replay vs --mode=live`

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Context Caching Strategy & Tools

Trap: Putting a dynamic {timestamp} inside static_instruction busts the cache on every single call!

static_instruction:
Stays cache-stable.

instruction:
Dynamic {state_key} placeholders append *after* the static block.

Context Management Tools:

include_contents='none':	Blinds the agent to history. Perfect for stateless classifiers/formatters.
EventsCompactionConfig:	Uses an LLM to summarize older event ranges into a compacted event.
ContextCacheConfig:	Reuses static prefixes on Gemini 2.0+ (requires App-level config).

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Agent Runtime

adk deploy agent_engine.
on GCP. Managed sessions, Memory Bank, code sandbox, Agent IAM Identity.

You manage: Agent code only.

Cloud Run

adk deploy cloud_run.
Scale-to-zero. Perfect for ambient (event-driven) agents triggered by Pub/Sub or Eventarc.

You manage: Container, DB, scaling.

GKE

adk deploy gke.
Heavy-duty. Network/residency control, GPU co-location, gVisor sandboxing.

You manage: Cluster, networking.

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DIAGNOSTIC TRAP & FIX QUICK REFERENCE

SYMPTOM / TRAP	SOLUTION / FIX
Trap: History lost on restart.	**Fix:** Swap InMemory for DatabaseSessionService.
Trap: LoopAgent never converged, but next step ran.	**Fix:** It exits silently on max_iterations. Check state flag in callbacks.
Trap: Parallel branches overwrite each other.	**Fix:** Branches must have distinct output_keys.
Trap: Early is_final_response() returns step 1.	**Fix:** Drain the run_async generator completely; every sub-agent emits it.
Trap: Setting threshold on tool_call_count_v1 crashes eval.	**Fix:** Built-in metrics are INFORMATIONAL. Define custom_metrics in test_config.json.
Trap: MCP tools fail on deployment start.	**Fix:** Define McpToolset synchronously at module level.

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Status
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Build the Agent, Not the Plumbing.
ADK supplies the memory, control flow, and governance so you can focus on the intelligence.
[Session End] // Infrastructure Deployed
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